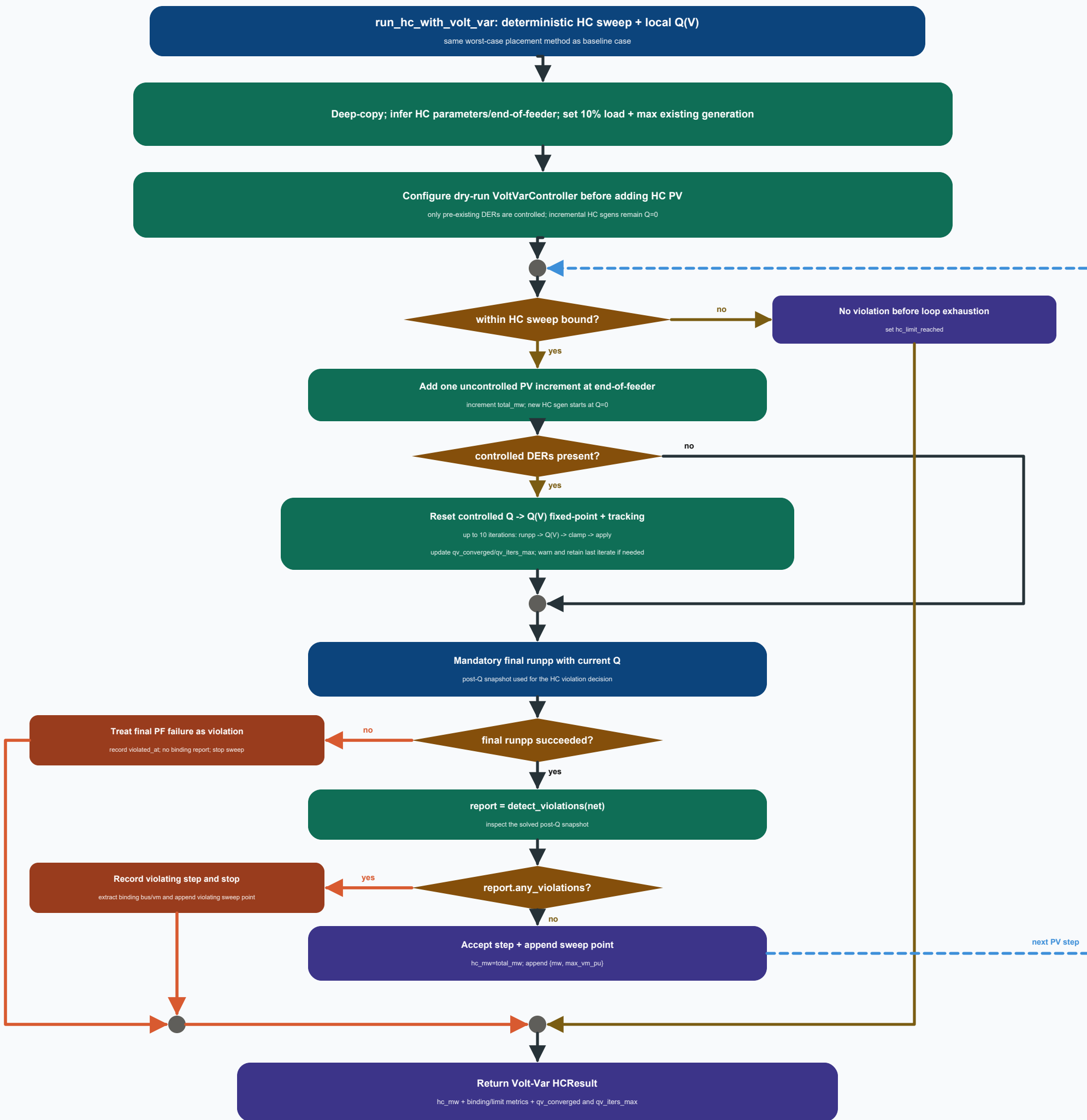




Volt-Var HC setup

Case B uses the same deep-copy, voltage-class, end-of-feeder and deterministic worst-case snapshot preparation as baseline HC.

VoltVarController is configured in dry-run mode before the HC sweep begins, using the same local Q(V) and inverter-limit logic as Scenario 4.

Because the controller resolves its sgen_indices at configuration time, PV capacity added later by the HC sweep is not Q-controlled.

Only the pre-existing controlled DER fleet supplies reactive support while incremental PV is tested at unity power factor.

Q(V) fixed-point inside each HC step

When controlled DERs exist, their Q is reset to zero and _qv_converge iterates up to MAX_QV_ITERS=10.

Each inner iteration runs a power flow, reads DER-bus voltages, computes Q(V), clamps to network/inverter limits and writes the new Q.

Convergence requires max|q_new-q_prev| < 1e-4 MVAR; non-convergence is logged and the last iterate is retained.

qv_converged and qv_iters_max are updated immediately after the inner function returns, before the final power flow.

The caller then always performs a final runpp; if no controlled DERs exist, the inner Q(V) loop is simply skipped.

Result semantics

A final-PF failure terminates the sweep without a voltage binding report; a solved violating snapshot records the binding bus and voltage.

Every clean snapshot becomes the latest hc_mw and contributes one {mw, max_vm_pu} point to sweep_curve.

Unlike baseline HC, Case B returns HCRresult only; its deep-copied working network remains internal to the analysis.

The benchmark runner compares Case B hc_mw against baseline hc_mw to quantify the hosting-capacity gain from the existing controllable DER fleet.